Fall 2025.

Physics 501. Methods of Theoretical Physics.
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Homework # 1.

Due to September 11th, 2025.

Read the corresponding chapters. If you forgot anything necessary from your core math courses, find the information in the textbook, online or whatever works for you. Solve the problems. It should be done in a very detailed and logical way. You need to understand every step of your solution.

Fourier series and orthogonal functions.

1. **Mark Pinsky, Section 0.3.** Problems 1, 2, 3, 7.
2. **Mark Pinsky, Section 1.1.** Problems 1, 3, 7, 9, 16 (a, b), 17 (a, b).
3. Consider a complex Fourier series. Prove that the Sturm-Liouville operator, $L = -\frac{d^2}{dx^2}$, is Hermitian on the set of imaginary exponents $\{e^{i\pi nx/L}\}$ and the interval *[-L; L]*.

Sturm-Liouville operator and orthogonal functions.

1. **Mark Pinsky, Section 1.6**. Problems 1, 3, 5.
2. **Arfken&Weber, Chapter 5.** Problems 5.1.9; 5.3.1, 5.3.2, 5.4.1-5.4.3.
3. **Arfken&Weber, Chapter 8.** Problems 8.2.1-8.2.6.
4. **Arfken&Weber, Chapter 12. Section 1.** Derive the Rodrigues formula (12.9). Apply it to the polynoms from Table 12.1.